Session 6. Measures of dispersion.

Pre course: Statistics.

Pablo Herrera G.



Hochschule
Bonn-Rhein-Sieg
University of Applied Sciences

Agenda

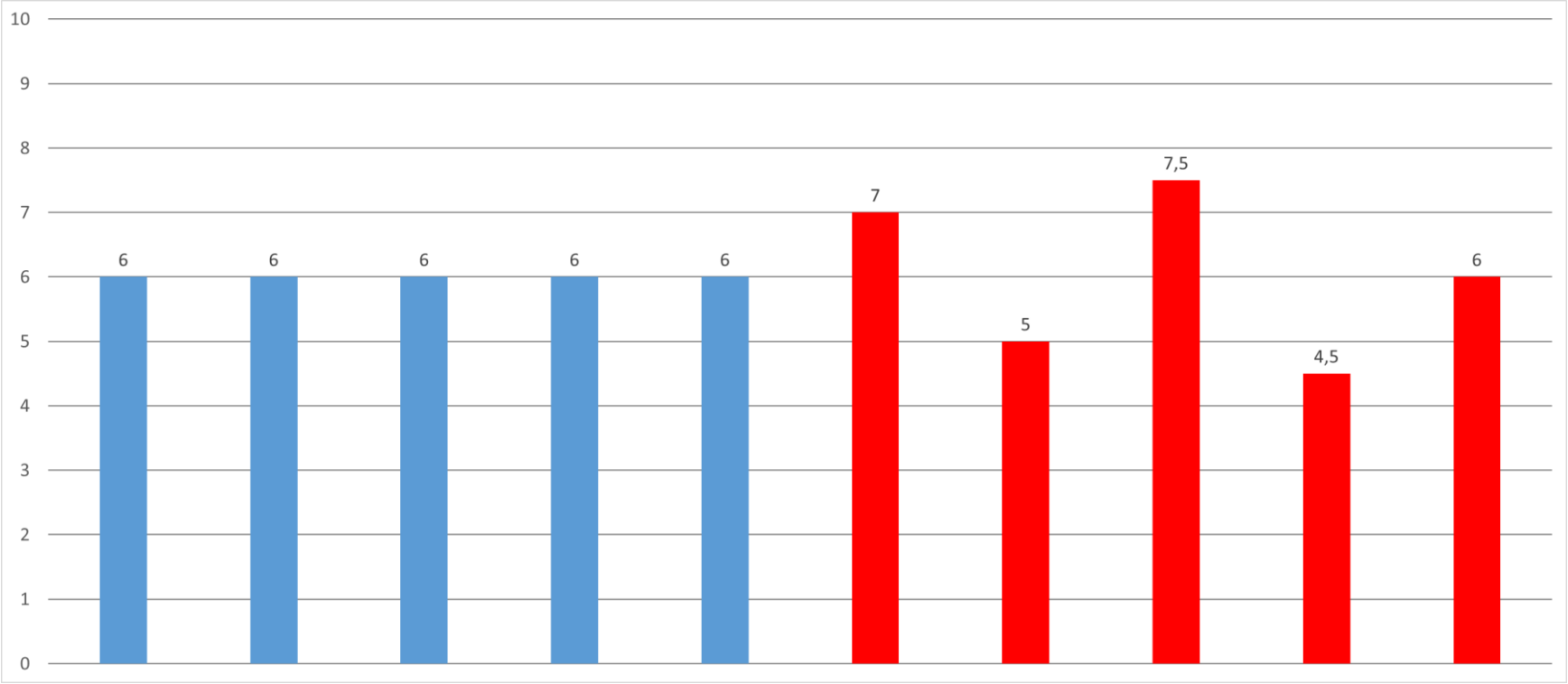
- **Motivation.**
- **Dispersion measures.**
- **Box plots.**
- **Effect of shifting and escalating data.**

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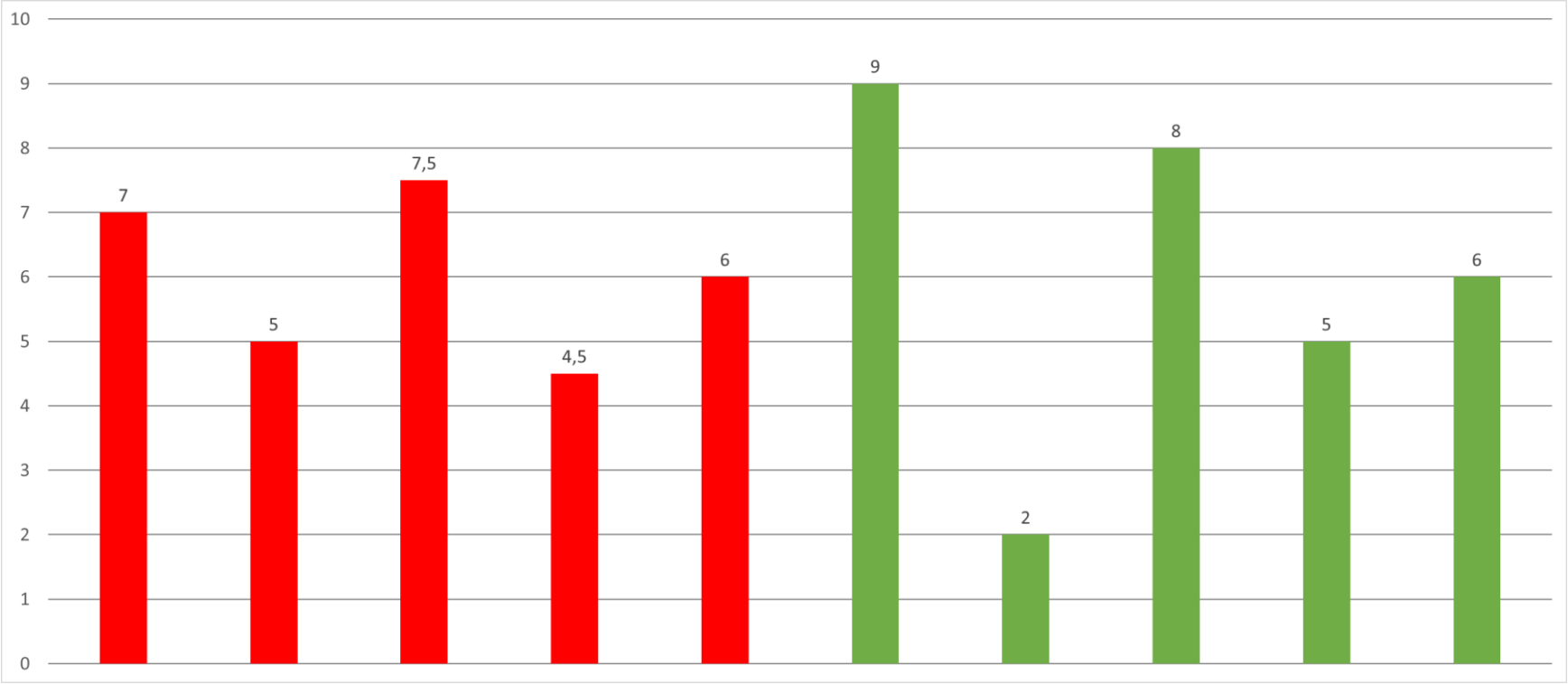
Motivation.

- Population Blue and Population Red.



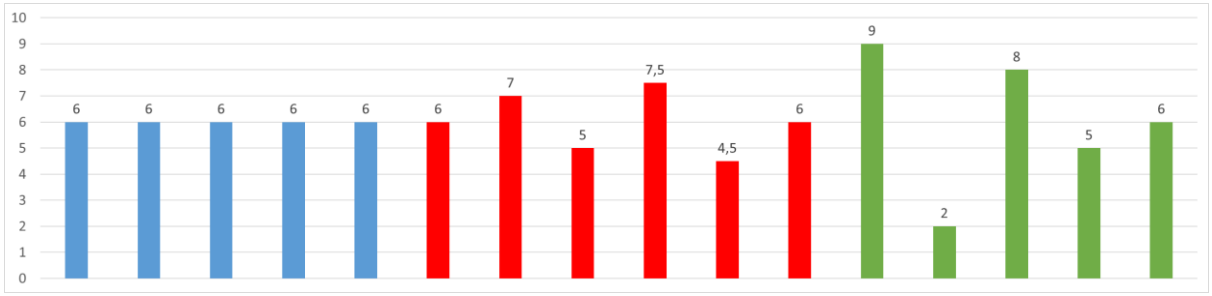
Motivation.

- Population Red and Population Green



Motivation.

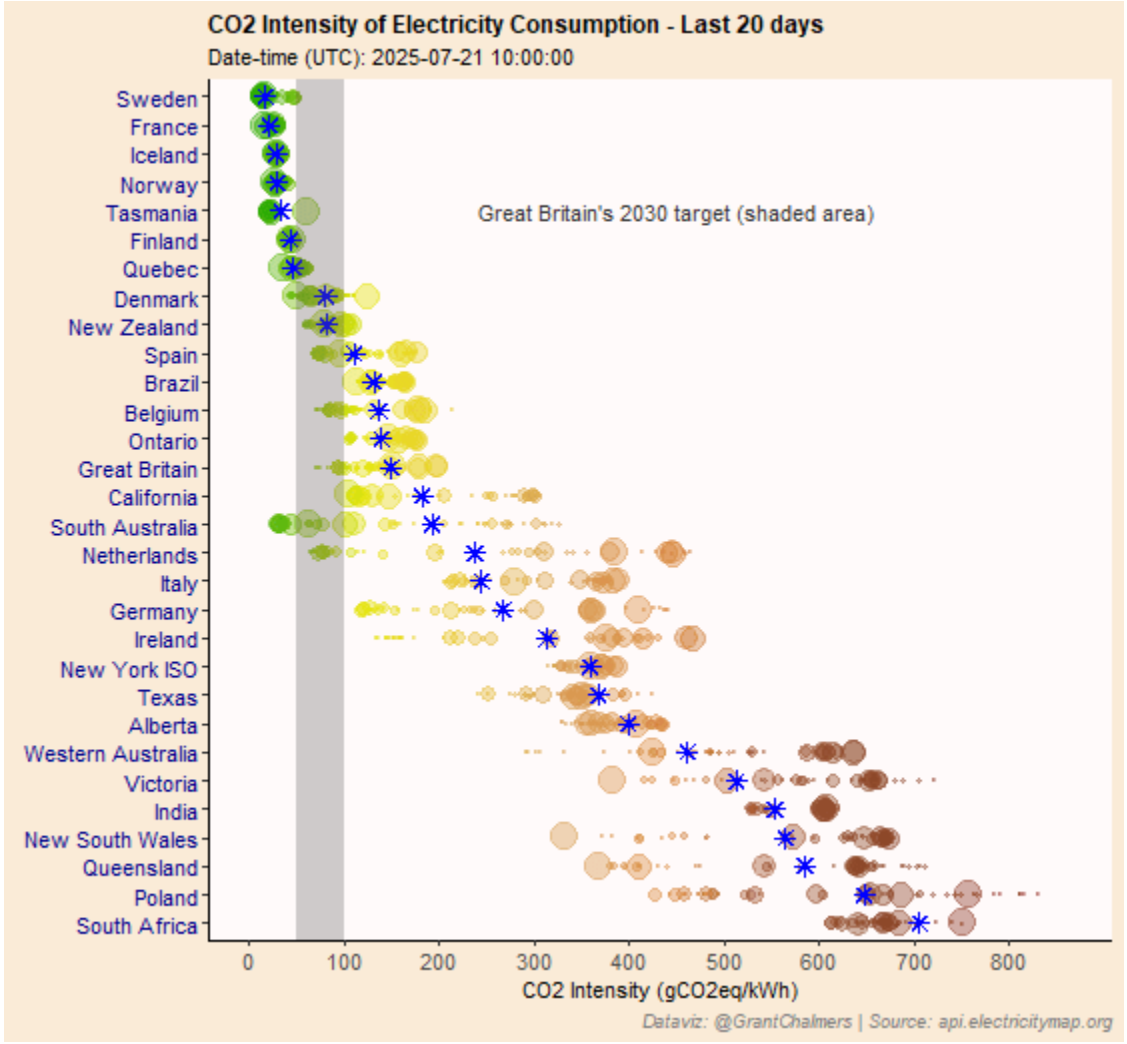
- Let's do the math.



Population Blue	Population Red	Population Green
6	7	9
6	5	2
6	7,5	8
6	4,5	5
6	6	6

Mean	6,00	6,00	6,00
Variance	0,00	1,63	7,50
Standard deviation	0,00	1,27	2,74

Motivation 2.



Source: https://www.linkedin.com/posts/grant-chalmers_rstats-electricity-dataviz-activity-7360570856982409219-xlrd?utm_source=share&utm_medium=member_desktop&rcm=ACoAABttqVQBmWXk3z1kCz_X22JwbAwLngTGUbk

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- Motivation.
- **Dispersion measures.**
- Box plots.
- Effect of shifting and escalating data.

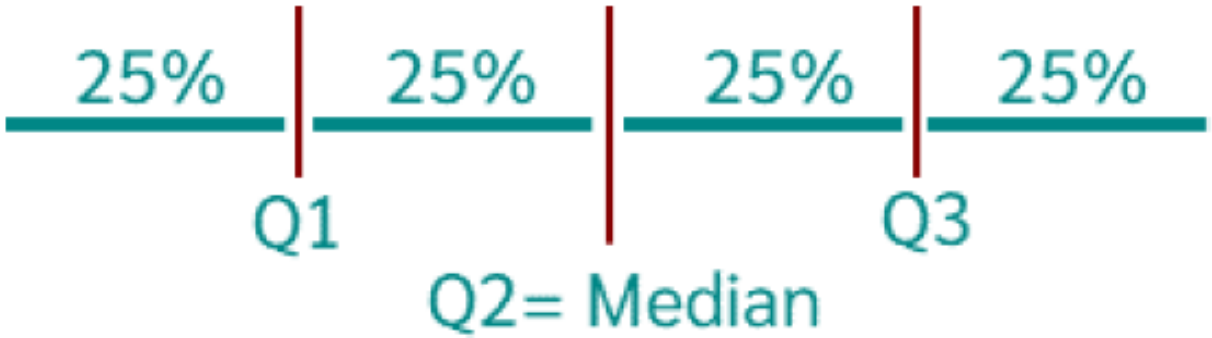
Dispersion measures.

- Interquartile range (IQR).
- Variance.
- Standard deviation.

Dispersion measures.

Interquartile range (IQR).

- Constructed with the measures of position (quantiles).
  - Quartiles, quintiles, deciles, percentiles.
  - For now, we will use the median half (can be inclusive or exclusive).
  - The are other methods too:
    - Method 1: (n+1)*p.
    - Method 2: (n−1)*p + 1.
- IQR = (Q3 – Q1) = (P.75 – p.25).



Dispersion measures.

Variance.

- Describes the average of the squared distances from the mean.

Standard deviation.

- It's a calculation that makes variance easier to interpret.
- It's the square root of the variance.

Dispersion measures.

Variance (var) or Standard deviation (sd)?

- Both are measures of dispersion, but the Std. Deviation facilitates interpretation.
- Suppose we are studying the dispersion of population income.
  - The variance will be expressed in EUR^2.
  - The standard deviation will be simply EUR.

Dispersion measures.

Variance (var) or Standard deviation (sd)?

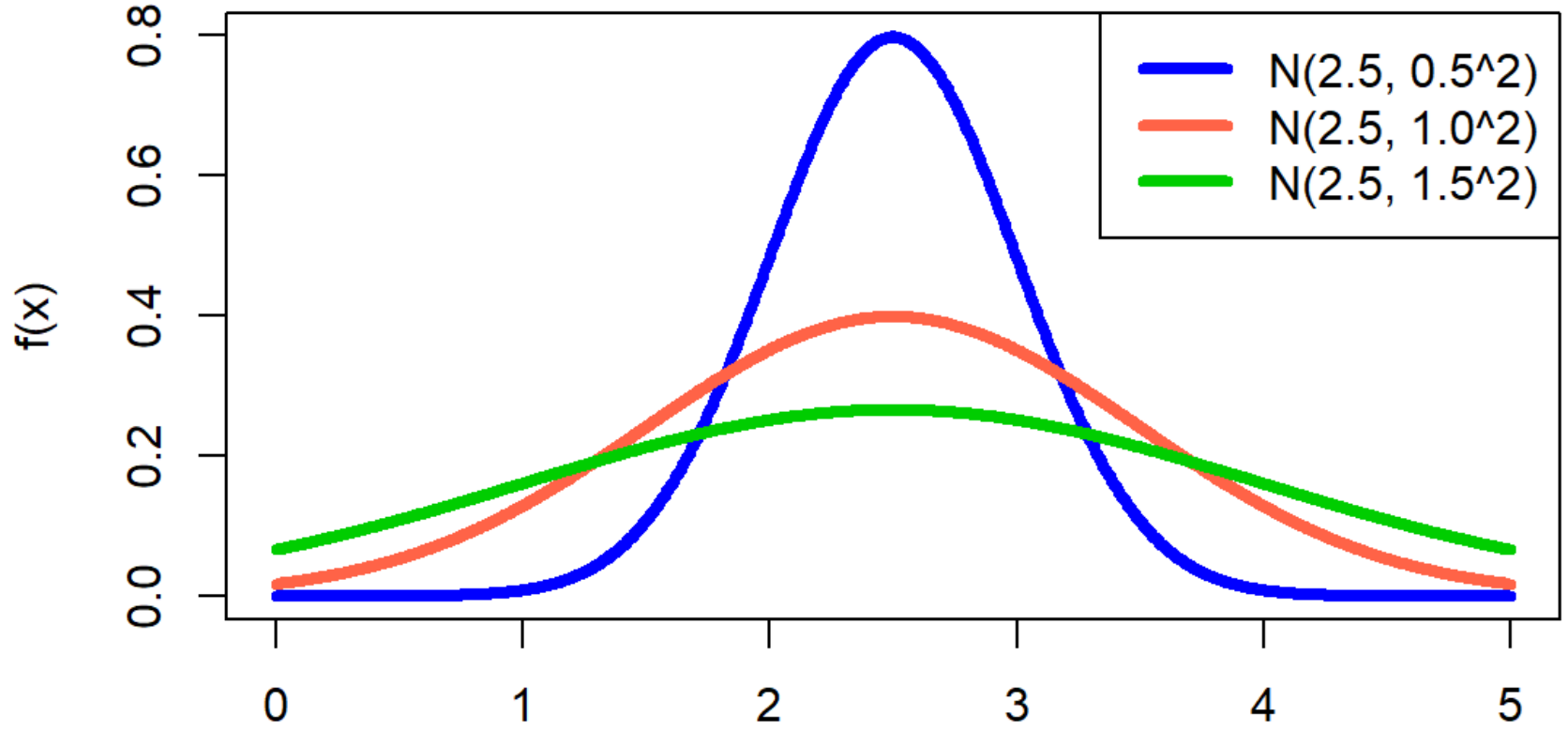
	Variance	Standard Deviation
Population	$\sigma^2 = \frac{\sum_{i=0}^{n}(x_i - \mu)^2}{n}$	$\sigma = \sqrt{\frac{\sum_{i=0}^{n}(x_i - \mu)^2}{n}}$
Sample	$s^2 = \frac{\sum_{i=0}^{n}(x_i - \bar{x})^2}{n-1}$	$s = \sqrt{\frac{\sum_{i=0}^{n}(x_i - \bar{x})^2}{n-1}}$

Dispersion measures.

- How does a higher or lower dispersion look like?

Dispersion measures.

- How does a higher or lower dispersion look like?
- Example with a normal distribution: N(mean, variance).

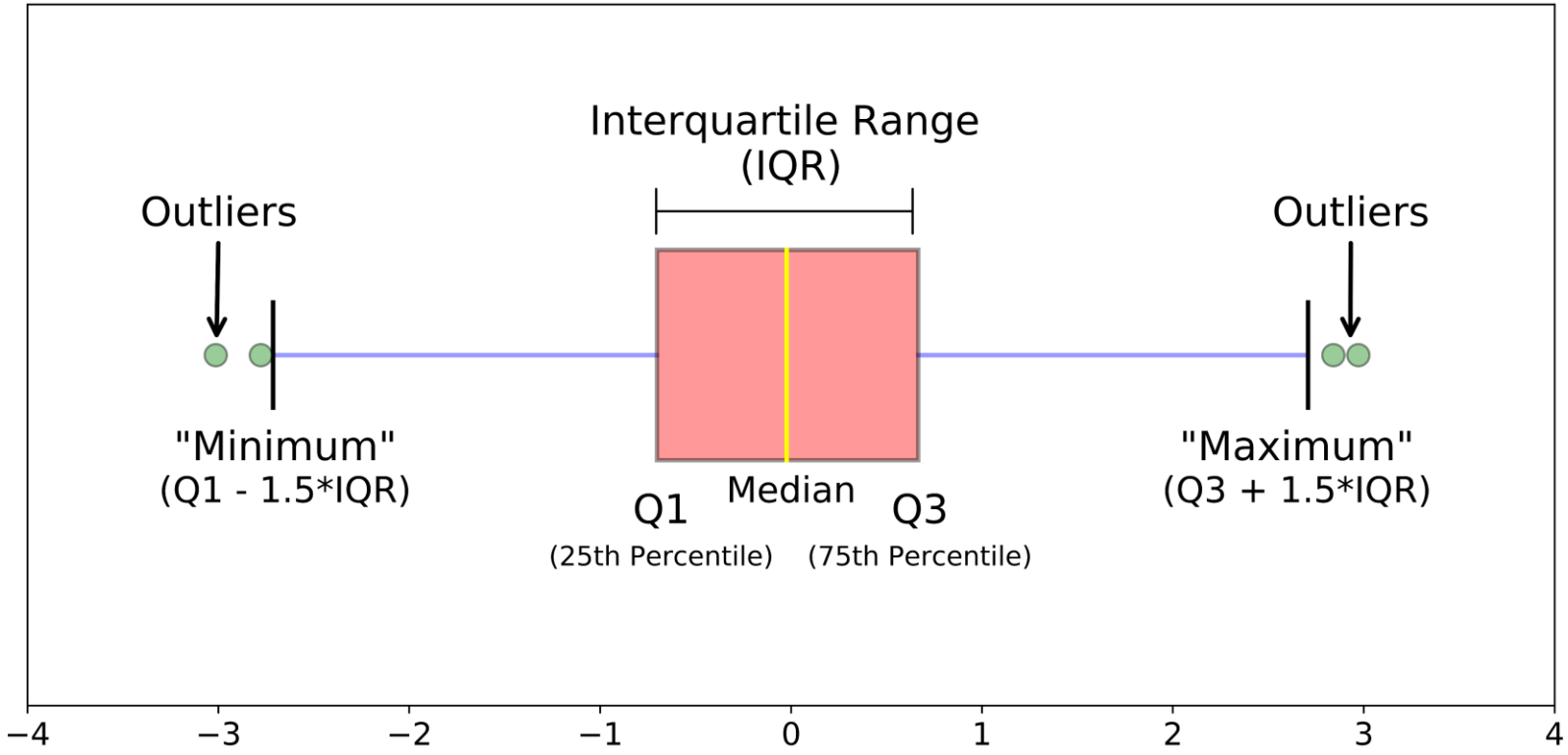


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- **Box plots.**
- Effect of shifting and escalating data.

Box plots.

- Visualization of a distribution's dispersion.
- The box is constructed with the Q1, Q2 and Q3. Whiskers may vary.



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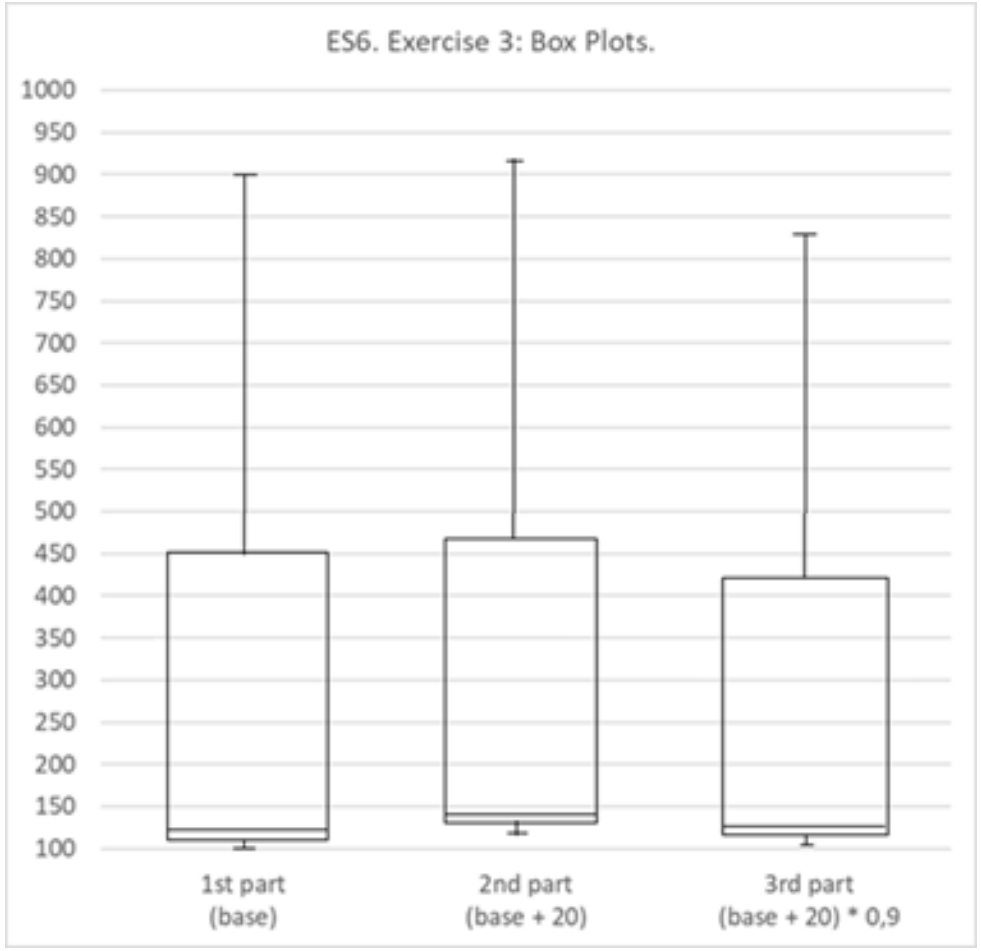
Effect of shifting and escalating data.

Based on exercises 1 and 2.

- Considering the central tendency and the dispersion measures
- What happens when we shift the data (2nd part).
- What happens when we rescale the data (3rd part)?

	1st part (base)	2nd part (base + 20)	3rd part (base + 20) * 0,9
	100	120	108
	110	130	117
	120	140	126
	450	470	423
	900	920	828
Mean	336	356	320
Percentile 25	110	130	117
Percentile 50 (or median)	120	140	126
Percentile 75	450	470	423
Variance	96.904	96.904	78.492
Standard deviation	311	311	280
Interquartile range (IQR)	340	340	306

Effect of shifting and escalating data.



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